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TECHNIQUES OF INTEGRATION



Due to the Fundamental Theorem of Calculus (FTC), we can integrate a function if we know an antiderivative, that is, an indefinite integral.

- We summarize the most important integrals we have learned so far, as follows.



FORMULAS OF INTEGRALS

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C \quad (n \neq -1)$$

$$\int \frac{1}{x} dx = \ln|x| + C$$

$$\int e^x dx = e^x + C$$

$$\int a^x dx = \frac{a^x}{\ln a} + C$$

$$\int \sin x \, dx = -\cos x + C$$

$$\int \cos x \, dx = \sin x + C$$

$$\int \sec^2 dx = \tan x + C$$

$$\int \csc^2 dx = -\cot x + C$$

$$\int \sec x \tan x \, dx = \sec x + C$$

$$\int \csc x \cot x \, dx = -\csc x + C$$

FORMULAS OF INTEGRALS

$$\int \sinh x\,dx = \cosh x + C \qquad \int \cosh x\,dx = \sinh x + C$$

$$\int \tan x\,dx = \ln|\sec x| + C \qquad \int \cot x\,dx = \ln|\sin x| + C$$

$$\int \frac{1}{x^2 + a^2}\,dx = \frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right) + C \qquad \int \frac{1}{\sqrt{a^2 - x^2}}dx = \sin^{-1}\left(\frac{x}{a}\right) + C$$

INTEGRATION BY PARTS

The Product Rule states that, if *f* and *g* are differentiable functions, then

$$\frac{d}{dx}\left[f(x)g(x)\right] = f(x)g'(x) + g(x)f'(x)$$

INTEGRATION BY PARTS

Formula 2

Thus, by the Substitution Rule, the formula for integration by parts becomes:

$$\int f(x)g'(x)\,dx = f(x)g(x) - \int g(x)f'(x)\,dx$$

$$\int u\,dv = uv - \int v\,du$$

Example:

Find $\int x \sin x \, dx$



E. g. 1—Solution 1

Using Formula 1, we have:

$$\int x \sin x \, dx = f(x)g(x) - \int g(x) f'(x) \, dx$$

$$= x(-\cos x) - \int (-\cos x) \, dx$$

$$= -x \cos x + \int \cos x \, dx$$

$$= -x \cos x + \sin x + C$$

- It's wise to check the answer by differentiating it.
- If we do so, we get $x \sin x$, as expected.

NOTE

If we had instead chosen $u = \sin x$ and $dv = x\,dx$, then $du = \cos x\,dx$ and $v = x^2/2$.

So, integration by parts gives:

$$\int x \sin x\,dx = (\sin x)\frac{x^2}{2} - \frac{1}{2}\int x^2 \cos dx$$

- Although this is true, $\int x^2 \cos x\,dx$ is a more difficult integral than the one we started with.

Remark:

Choose *u* as the type of function that appears first in **LIPTE**

L logarithmic(ln x)

I Inverse trigonometric

P Polynomial (x^n)

T Trigonometric (sinx…..)

E Exponential (e^x)

Example:

Find: $\int \ln x \, dx$

L logarithmic(ln x)
I Inverse trigonometric
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Let $u = \ln x \qquad dv = dx$

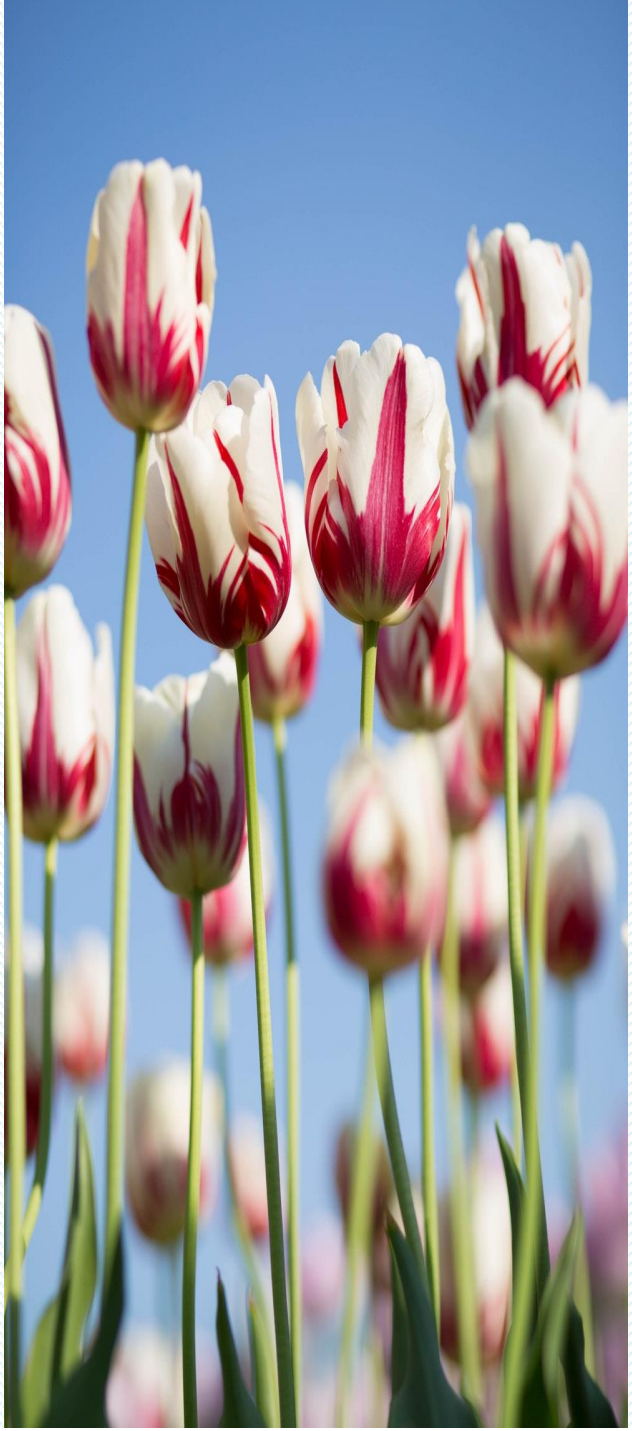
$$du = \frac{1}{x}dx \qquad v = x$$

Integrating by parts, we get:

$$\int \ln x\,dx = x\ln x - \int x\frac{dx}{x}$$

$$= x\ln x - \int dx$$

$$= x\ln x - x + C$$



Remark:

Integration by parts is effective in this example because the derivative of the function $f(x) = \ln x$ is simpler than f.



Example 3

Find $\int t^2 e^t dt$

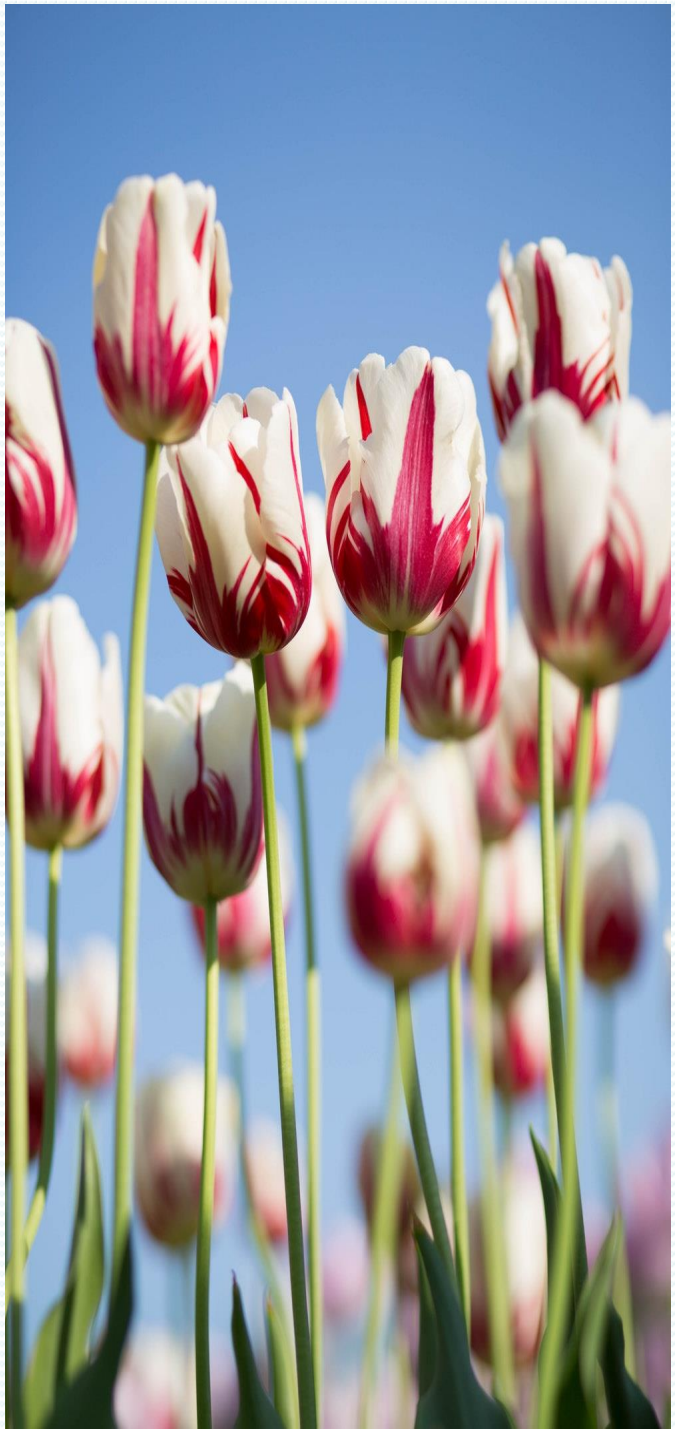
L logarithmic(ln x)

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E. g. 3—Equation 3

So, we choose $u = t^2 \quad dv = e^t dt$

Then, $du = 2t\,dt \qquad v = e^t$

Integration by parts gives:

$$\int t^2 e^t dt = t^2 e^t - 2\int te^t dt$$

Note that:

The integral that we obtained, $\int te^t dt$,

is simpler than the original integral.

However, it is still not obvious.

- So, we use integration by parts a second time.



This time, we choose

$$u = t \text{ and } dv = e^t dt$$

- Then, $du = dt$, $v = e^t$.
- So, $\int te^t dt = te^t - \int e^t dt - te^t - e^t + C$

Putting this in Equation 3, we get

$$\int t^2 e^t dt = t^2 e^t - 2\int te^t dt$$

$$= t^2 e^t - 2(te^t - e^t + C)$$

$$= t^2 e^t - 2te^t - 2e^t + C_1$$

where $C_1 = -2C$

INTEGRATION BY PARTS

Example 4

Evaluate $\int e^x \sin x \, dx$

- e^x does not become simpler when differentiated.
- Neither does sin x become simpler.

L logarithmic(ln x)

I Inverse trigonometric

P Polynomial (x^n)

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E Exponential (e^x)

E. g. 4—Equation 4

Nevertheless, we try choosing

$$u = e^x \text{ and } dv = \sin x$$

- Then, $du = e^x\,dx$ and $v = -\cos x$.

Example 4

So, integration by parts gives:

$$\int e^x \sin x \, dx = -e^x \cos x + \int e^x \cos x \, dx$$

Example 4

The integral we have obtained, $\int e^x \cos x \, dx$, is no simpler than the original one.

- At least, it's no more difficult.
- Having had success in the preceding example integrating by parts twice, we do it again.

E. g. 4—Equation 5

This time, we use

$$u = e^x \text{ and } dv = \cos x\, dx$$

Then, $du = e^x\, dx$, $v = \sin x$, and

$$\int e^x \cos x\, dx = e^x \sin x - \int e^x \sin x\, dx$$

Example 4

At first glance, it appears as if we have accomplished nothing.

- We have arrived at $\int e^x \sin x \, dx$, which is where we started.

Example 4

However, if we put the expression for $\int e^x \cos x \, dx$ from Equation 5 into Equation 4, we get:

$$\int e^x \sin x \, dx = -e^x \cos x + e^x \sin x - \int e^x \sin x \, dx$$

- This can be regarded as an equation to be solved for the unknown integral.

Example 4

Adding to both sides $\int e^x \sin x \, dx$, we obtain:

$$2\int e^x \sin x \, dx = -e^x \cos x + e^x \sin x$$

Example 4

Dividing by 2 and adding the constant of integration, we get:

$$\int e^x \sin x \, dx = \tfrac{1}{2} e^x (\sin x - \cos x) + C$$

INTEGRATION BY PARTS

If we combine the formula for integration by parts with Part 2 of the FTC (FTC2), we can evaluate definite integrals by parts.

INTEGRATION BY PARTS

Formula 6

Evaluating both sides of Formula 1 between *a* and *b*, assuming *f'* and *g'* are continuous, and using the FTC, we obtain:

$$\int_a^b f(x)g'(x)\,dx = f(x)g(x)\Big]_a^b - \int_a^b g(x)f'(x)\,dx$$

INTEGRATION BY PARTS

Example 5

Calculate $\int_0^1 \tan^{-1} x \, dx$

- Let $u = \tan^{-1} x \qquad dv = dx$
- Then, $du = \dfrac{dx}{1+x^2} \qquad v = x$

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Example 5

So, Formula 6 gives:

$$\int_0^1 \tan^{-1} x\,dx = x\tan^{-1} x\Big]_0^1 - \int_0^1 \frac{x}{1+x^2}\,dx$$

$$= 1\cdot\tan^{-1}1 - 0\cdot\tan^{-1}0 - \int_0^1 \frac{x}{1+x^2}\,dx$$

$$= \frac{\pi}{4} - \int_0^1 \frac{x}{1+x^2}\,dx$$

Example 5

To evaluate this integral, we use the substitution $t = 1 + x^2$ (since u has another meaning in this example).

- Then, $dt = 2x\,dx$.
- So, $x\,dx = \frac{1}{2}\,dt$.

Example 5

When $x = 0$, $t = 1$, and when $x = 1$, $t = 2$.

Hence, $$\int_0^1 \frac{x}{1+x^2}\,dx = \tfrac{1}{2}\int_1^2 \frac{dt}{t}$$

$$= \tfrac{1}{2}\ln|t|\Big]_1^2$$

$$= \tfrac{1}{2}(\ln 2 - \ln 1)$$

$$= \tfrac{1}{2}\ln 2$$

Example 5

Therefore,

$$\int_0^1 \tan^{-1} x \, dx = \frac{\pi}{4} - \int_0^1 \frac{x}{1+x^2} \, dx$$

$$= \frac{\pi}{4} - \frac{\ln 2}{2}$$